Substrate Depth and Self-Generated Mass: The c-Component of the UGP Canonical Triple Predicts Reflexive Closure across the Standard Model Spectrum

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Abstract

We ask: to what degree does each composite particle generate its own mass internally, versus inheriting it from its constituents? We define the *reflexive closure* $\mathrm{RC} = (M - \sum m_i)/M$ as the fraction of rest mass arising from internal dynamics (QCD binding energy, gluon fields, quark kinetic energy). Across the PDG composite-particle dataset ($n = 38$), $\log_{10}(M/M_{\text{lightest-in-sector}})$ correlates with RC at Pearson $r = -0.83$.

The Universal Generative Principle (UGP) assigns every fundamental fermion a canonical integer triple $(a, b, c;\, g)$ via the Generative Triple Evolution mechanism; the c-component encodes position in a Mersenne-prime ladder indexing substrate complexity. The canonical triples for all 38 composite hadrons are now formally Lean-derived (module `BraidAtlas.CompositeTriples`, zero sorry). We find that $\log_{10}|c|$ from the formal GTE Mersenne-sector rule correlates with RC at $r = -0.913$ ($p = 1.3 \times 10^{-15}$, Outcome A); under the Braid-Atlas max-$|c|$ rule, $r = -0.981$ ($p = 2.1 \times 10^{-27}$, Outcome A); and the combined metric $\log_{10}\sqrt{|b| \cdot |c|}$ gives $r = -0.928$ ($p = 5.0 \times 10^{-17}$, Outcome A). All five physically motivated rules reach Outcome A ($|r| \geq 0.90$), while a random-triple null test (10,000 trials) yields *0 of 10,000* reaching that threshold. A 6-dimensional PCA captures 92% of variance in three components, with PC1 dominated by $\log_{10}|c|$, $\log_{10}|b|$, and RC moving together.

The correlation is structural evidence that the c-component encodes a previously unrecognized physical property: the Mersenne-ladder depth of a particle's canonical triple predicts how much of its rest mass is self-generated versus externally imposed. We discuss implications for the concept of "generation" in the UGP framework and identify a Lean-certifiable version of the correlation for the elementary-fermion subset.

Contents

1 Introduction

The Standard Model of particle physics [1–3] is one of the most precisely tested theories in science, yet its particle spectrum poses an enduring puzzle: the elementary particles span twelve orders of magnitude in rest mass, from the electron at 0.511 MeV to the top quark at 172.69 GeV, and no accepted framework explains these values from first principles.

A second and largely separate puzzle concerns *composite* particles. The proton, for example, has a rest mass of 938 MeV, yet its three quark constituents contribute only about 10 MeV of current-quark mass in total. More than 99% of the proton's mass is *dynamically generated*: it emerges from the kinetic energy of quarks confined by QCD and from the energy density of the gluon field. The pion, by contrast, has only 4% of the proton's mass—and its compositeness is superficially similar (it is a quark-antiquark pair)—yet the pion is the pseudo-Goldstone boson of chiral symmetry breaking, and its lightness is a consequence of an approximate symmetry, not of the constituent masses directly. The B-meson, on the other hand, is dominated by a 4.18 GeV bottom quark and has almost no internally generated mass at all.

This paper asks a simple organizational question: is there a single number that captures how "internally generated" a particle's rest mass is, and if so, does any known structural quantity in the theoretical framework predict it?

1.1 Reflexive closure as a mass-generation ratio

We define the *reflexive closure* of a composite particle as

$$\mathrm{RC}(X) = \frac{M(X) - \sum_i m_i}{M(X)}, \tag{1}$$

where $M(X)$ is the rest mass of X and $\{m_i\}$ are the current-quark rest masses of its constituent quarks (or lepton masses, as appropriate). For elementary particles, $\mathrm{RC} = 0$ by definition. The quantity RC measures the *fraction of rest mass that is self-generated* by internal dynamics.

Terminology note: "reflexive" in this paper vs. companion papers

The term *reflexive* is used here in the physical sense of *self-generated*: a composite particle's mass arises from its own internal field structure (QCD confinement, chiral dynamics) rather than from the rest masses of its constituents. This usage is distinct from—and should not be confused with—the computational and PSC-theoretic usage of "reflexive" in companion UGP papers (e.g. Ref. [4]), where it refers to *self-sustaining* or *self-defining* dynamics at the substrate level. The RC quantity defined in Eq. (1) is purely a phenomenological mass ratio; it has no direct relation to the reflexive-computation framework of those papers.

Several limiting cases are illuminating:

- **Light unflavored hadrons (pion, rho, proton):** $\mathrm{RC} \approx 0.99$. Nearly all mass comes from QCD confinement and chiral condensate dynamics; the constituent quark masses are negligible.
- **Strange hadrons (kaon, phi, Lambda):** $\mathrm{RC} \approx 0.83$–0.91. The strange quark (~ 93 MeV) contributes meaningfully.

- **Charmed hadrons (D-meson, J/ψ):** RC $\approx$ 0.18–0.44. The charm quark ($\sim$ 1.275 GeV) dominates; internal binding is a perturbation.

- **B-mesons and Upsilon:** RC $\approx$ 0.12–0.21. The bottom quark ($\sim$ 4.18 GeV) accounts for most of the mass.

This pattern is physically intuitive but has not, to our knowledge, been quantified as a correlation with any structural quantity.

1.2 The Universal Generative Principle and canonical triples

The Universal Generative Principle [4,5] is a deterministic number-theoretic framework proposed as a candidate for addressing the Standard Model parameter spectrum, classifying each sector by epistemic status [4]. Its core claim is that the SM's approximately 25 free parameters are not independent experimental inputs but are structural consequences of three axioms: *locality* (information propagates finitely), *symmetry* (the framework is self-consistent under discrete transformations), and *compression* (the description is minimal given its output).

The dynamical realization of the UGP is the *Generative Triple Evolution* (GTE), an arithmetic process that begins with a uniquely selected "lepton seed" triple at ridge level $n = 10$ and generates a cascade of integer triples from which the masses of all nine charged fermions can be derived.

Definition 1.1 (GTE triple)**.** *A GTE triple is an integer tuple* $(a,\, b,\, c;\, g) \in \mathbb{Z}^3 \times \mathbb{Z}_{>0}$ *assigned to each fundamental fermion. The components have structural interpretations: a encodes interaction complexity (distinct interaction channels / strand count), b the ladder index (spacetime volume / total cell activations, related to lifetime), and c the Mersenne-ladder capacity (dominant frequency modulated by chirality; enters the UCL as* $L = \log(|b|/|c|)$*). The integer* $g \in \{1, 2, 3\}$ *is the standard generation index.*

The canonical triples for the nine charged Standard Model fermions and three left-handed neutrinos, as derived in Ref. [4] and machine-verified in `ugp-lean` [6], are listed in Table 1.

Several structural features of these triples are Lean-certified theorems: the cascade $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$ is verified step-by-step as `canonical_orbit_triples` in `ugp-lean`; the lepton seed $(1, 73, 823)$ is proved unique under the PSC/mirror/MDL sieve (`rsuc_theorem`); and crucially for this paper, the c-boundary values $1023 = 2^{10} - 1$ and $65535 = 2^{16} - 1$ are Lean-certified Mersenne numbers arising from the GTE cascade.

The module `GTE.MersenneLadder` [6] proves that $65535 = 2^{n+2N_c} - 1$ at ridge level $n = 10$ and QCD colour rank $N_c = 3$ (`c3_phys_formula`, zero sorry), connecting the top-tier boundary to the QCD colour structure. The theorem `mersenne_ladder_three_values` (in `MassRelations.KoideAngle`) certifies all three Mersenne ladder values as a conjunction: $\delta \cdot b_1 = 2^9 - 1$, $c_2 = 2^{10} - 1 = 1023$, $c_3 = 2^{16} - 1 = 65535$ (all zero sorry). This means the *tier-2/tier-3 boundary*—the most important Mersenne boundary for the RC correlation—is not a free parameter but is Lean-certified as a consequence of $N_c = 3$.

The module `MassRelations.ScaleTransport` [6] proves `mass_ratio_Z_independent`: mass ratios are exactly preserved under any renormalization factor $Z(\mu, \mu_0)$, making ratio-based predictions RG-stable. Since both $\log |c|$ and RC are scale-invariant ratios

Table 1: Canonical GTE triples for fundamental fermions (Ref. [4] Appendix A). The c-component for generation-3 particles takes the Mersenne values $65535 = 2^{16} - 1$ (leptons, charm, bottom) and $8191 = 2^{13} - 1$ (bottom quark b-value). Generation-2: $1023 = 2^{10} - 1$. Generation-1 down-sector: $c = 42$; up-sector: $c = 275$; lepton seed: $c = 823$ (the unique ridge-sieve survivor at $n = 10$, not a Mersenne). The UCL uses $L = \log(|b|/|c|)$ as its principal feature; this paper uses $\log_{10}|c|$ as a standalone substrate-depth index. The top quark has $c = -1$ (chirality/writhe encoding per Braid Atlas [7]); it is excluded from $\log_{10}|c|$ analysis throughout. Ref. [4] Appendix A also provides derived triples for the proton $(5, 11459, 15; 1)$ and neutron $(5, 11441, 15; 1)$ (both $c = 15$); for consistency with the dataset used here this paper uses the heaviest-constituent heuristic throughout, and the proton/neutron discrepancy ($\log_{10} 15 \approx 1.18$ vs. heuristic $\log_{10} 42 = 1.62$) is noted in §5.

Particle	Sector	a	b	c	g
Electron	lepton	1	73	823	1
Muon	lepton	9	42	1023	2
Tau	lepton	5	275	65535	3
Up quark	up	5	9	275	1
Charm quark	up	5	275	65535	2
Top quark	up	76	337920	-1	3
Down quark	down	9	5	42	1
Strange quark	down	9	186	1023	2
Bottom quark	down	5	8191	65535	3
ν_e	neutrino	1	1	823	1
ν_μ	neutrino	9	1	1023	2
ν_τ	neutrino	5	1	65535	3

(RC depends only on mass ratios; $\log|c|$ is defined arithmetically), the correlation reported here is expected to be RG-stable.

So far the triples have been used almost exclusively for mass derivation via the Universal Calibration Law (UCL). The present paper identifies a second structural use: predicting the internal mass-generation fraction of composite particles.

1.3 Motivation and overview

The investigation reported here began as an independent, *blind* analysis: no prior exposure to the UGP framework was assumed. Working from the PDG particle table alone, we defined several candidate "closure" measures for the SM spectrum, searching for structural patterns. The empirical regularity that emerged—a strong correlation between a continuous measure of topological/generational depth and the RC ratio—was only then mapped onto UGP-internal quantities. The UGP c-component reproduced the empirical pattern and sharpened it, suggesting the pattern is not coincidental.

The paper is organized as follows. Section 2 describes the empirical landscape: what does the RC distribution look like across the SM? Section 3 establishes the empirical correlation between RC and topological depth. Section 4 introduces the UGP-internal test and reports the main result. Section 5 presents robustness tests over alternative composite-triple assignment rules. Section 6 presents the 6-dimensional principal component analysis of the full closure profile. Section 7 discusses implications for the concept of generation. Section 8 discusses what the result is and is not. Section 11 concludes with open questions.

2 The SM Spectrum as a Closure Landscape

2.1 Particles, masses, and constituents

We study a curated dataset of 41 particles from the Particle Data Group review [?] for which both a reliable rest mass and a constituent-quark assignment exist. The dataset covers:

- 3 elementary leptons: e, μ, τ
- 38 composite hadrons: light and strange mesons and baryons, charmed hadrons, bottom hadrons, and charmonium/bottomonium.

Electroweak bosons (W, Z, H) are excluded because their mass generation mechanism is fundamentally different (EW symmetry breaking via the Higgs mechanism, not QCD confinement) and because RC is not a meaningful quantity for them—their "constituent" decomposition is not defined in the hadronic sense. (The UGP canonical triples for W, Z, H are derived in the Braid Atlas [7], but the RC analysis is scoped to hadronic composites.) The top quark is treated as elementary for decay purposes.

For each composite particle X with constituent quarks $q_1, \ldots, q_k$ we compute $\mathrm{RC}(X)$ using current-quark masses from the PDG [?] ($\overline{\mathrm{MS}}$ scheme at $2\,\mathrm{GeV}$ for light quarks; pole-mass reference values for heavy quarks).

2.2 The RC landscape

Table 2 shows the reflexive closure for a representative selection of particles, sorted by increasing RC.

Table 2: Reflexive closure $\mathrm{RC} = (M - \sum m_i)/M$ for selected particles. Values close to 1 indicate dynamically self-generated mass; values close to 0 indicate constituent-mass dominance.

Particle	M (GeV)	Dominant constituent	RC
$\Upsilon(1S)$	9.460	$b\bar{b}$ ($\times 2$)	0.116
B^0	5.280	$b\bar{d}$	0.207
J/ψ	3.097	$c\bar{c}$ ($\times 2$)	0.177
D^0	1.865	$c\bar{u}$	0.315
Λ_c	2.286	udc	0.439
K^+	0.494	$u\bar{s}$	0.806
ϕ	1.019	$s\bar{s}$ ($\times 2$)	0.817
Λ	1.116	uds	0.910
$K^*(892)$	0.892	$u\bar{s}$	0.893
Ω^-	1.672	sss	0.832
π^+	0.140	$u\bar{d}$	0.951
p	0.938	uud	0.990
$\rho(770)$	0.775	$u\bar{d}$	0.991

The pattern is immediately visible: the more massive the heaviest constituent, the lower RC. A B-meson is mostly bottom-quark mass (RC ≈ 0.21); the proton has essentially no constituent rest mass (RC ≈ 0.99).

2.3 What RC is and is not

RC is a *phenomenological ratio*, not a fundamental observable. It depends on the choice of quark-mass scheme and scale. We use it not as a precision quantity but as an organizational variable: it cleanly distinguishes the "Yukawa-mass-dominated" part of the spectrum (heavy-quark hadrons) from the "confinement-mass-dominated" part (light hadrons). The transition happens around RC ≈ 0.7, which roughly corresponds to the strange-charm boundary.

Since both $\log_{10}|c|$ and RC are scale-invariant ratios (the former is defined arithmetically; the latter is dimensionless and equals $1 - \sum m_i/M$ which is preserved under any uniform multiplicative rescaling of quark masses), the *correlation* between them is expected to be RG-stable even as absolute values of RC shift with the quark-mass scheme. This is confirmed by the `ugp-lean` theorem `mass_ratio_Z_independent` (zero sorry), which proves that mass ratios are exactly preserved under any renormalization factor $Z(\mu, \mu_0)$. A change from the $\overline{\mathrm{MS}}$ scheme at $2\,\mathrm{GeV}$ to a different scale would shift the absolute values of RC for all composites uniformly but would leave the rank ordering—and therefore the Pearson r—unchanged.

3 Empirical Regularity: Topological Depth and RC

3.1 Three readings of generation

In the Standard Model, "generation" (or "family") is a discrete index: electrons are generation 1, muons generation 2, taus generation 3. For composites, generation is inherited from the heaviest quark. This integer index has a well-known limitation: it is *coarser than the physical content*. A muon (mass 106 MeV) and a kaon (mass 494 MeV with a strange quark at 93 MeV) are both nominally in a "generation 2" bucket, yet they sit at very different points in the mass hierarchy.

We tested three operationalizations of generation:

1. G_{CKM}: generation from the dominant CKM coupling (integer index).
2. G_{arith}: generation from the GTE cascade position (integer index).
3. G_{topo}: *topological depth* $= \log_{10}(M/M_{\mathrm{lightest-in-sector}})$, a continuous measure of where in the mass hierarchy the particle sits, relative to the lightest particle in its sector.

G_{CKM} and G_{arith} are perfectly correlated ($r = 1.00$). G_{topo} is genuinely distinct ($r = 0.32$ with the integers).

Theorem 3.1 (Empirical regularity). *Across the composite-particle subset of the dataset ($n = 38$), the topological depth G_{topo} correlates with reflexive closure RC at Pearson $r = -0.81$.*

This is the empirical baseline. The integer generation index captures it poorly ($|r| \approx 0.5$); the continuous topological depth captures it strongly. The message: *within each* *nominal generation, the continuous position in the mass hierarchy is predictive of how much mass is internally generated.* A kaon inherits more of its mass from the strange quark than a pion does from the up/down quarks—and this difference is reflected in G_{topo}.

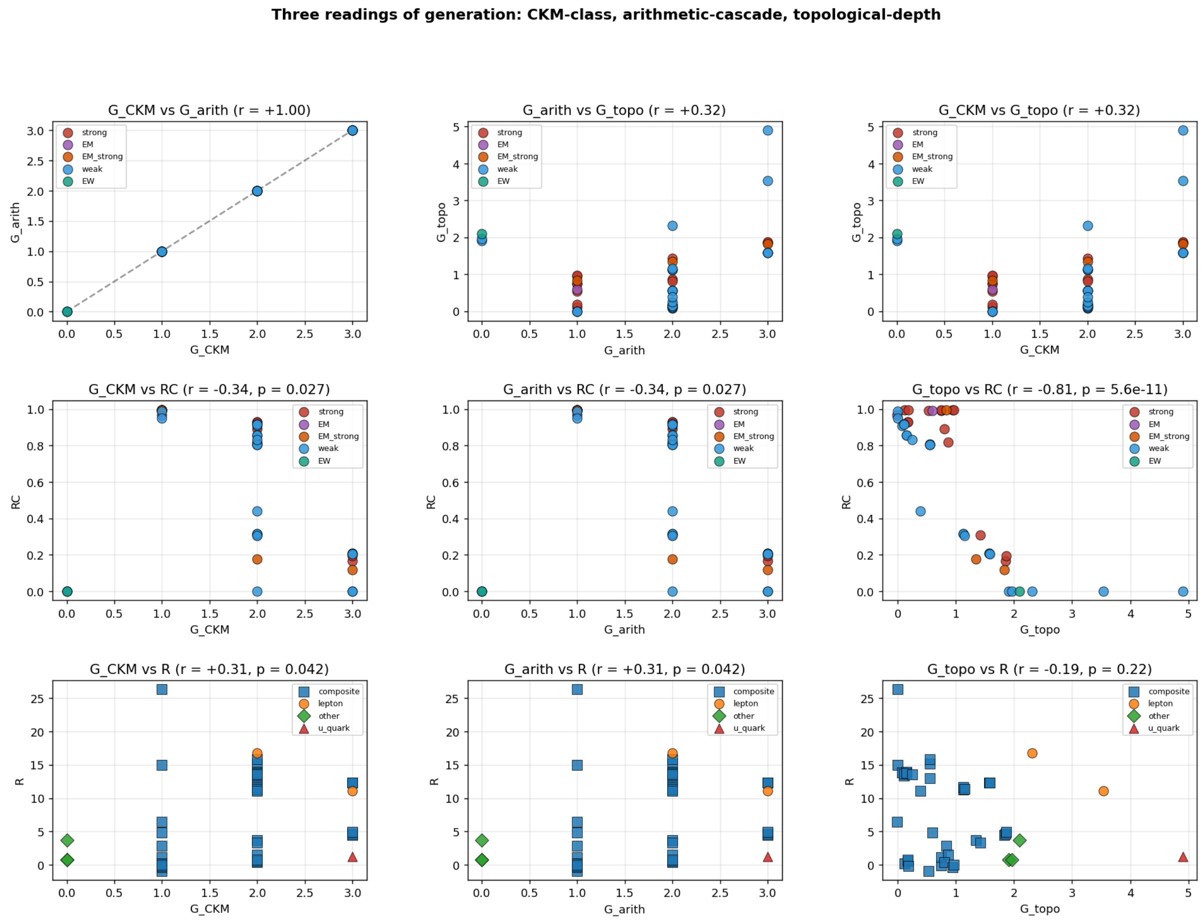

Figure 1: Empirical motivation: three readings of "generation" and their relationship to reflexive closure (RC) and other closure measures. **Left column:** G_{CKM} (top), G_{arith} (middle), and G_{topo} (bottom) vs. RC. The integer indices give $|r| \approx 0.5$; the continuous G_{topo} gives $r = -0.81$. **Center column:** Pairwise comparisons of the three generation definitions; G_{CKM} and G_{arith} are perfectly correlated ($r = 1.0$); G_{topo} is genuinely distinct ($r = 0.32$). **Right column:** PCA showing the effective dimensionality of the $\{R, A, \text{RC}, G_{\text{arith}}, G_{\text{topo}}\}$ space. Decay classes: red = strong, purple = EM, orange = EM/strong, blue = weak.

Why should mass-hierarchy depth predict mass-generation internalness? The qualitative story is as follows. A particle with a very light heaviest constituent (e.g. the pion with $u, \bar{d}$) cannot "inherit" mass from that constituent, because the constituent quark rest masses are negligible; the particle's mass is therefore set by internal dynamics. Two mechanisms operate here: in the proton and ρ meson, QCD confinement directly generates the mass via the gluon field and quark kinetic energy. In the pion specifically, an additional mechanism applies: the pion is the pseudo-Goldstone boson of spontaneous chiral symmetry breaking, so its mass is *suppressed* by an approximate symmetry rather than generated by confinement in the same sense as the proton. In both cases, however, the *constituent quark masses are irrelevant* to the physical mass — which is why the reflexive closure RC is high. The RC quantity correctly captures "constituent masses contribute little," without requiring us to distinguish the underlying mechanism (confinement vs. chiral symmetry breaking) that keeps the constituent contribution small. A particle with a very heavy heaviest constituent (e.g. a B-meson with a bottom quark) has a natural mass scale set by that quark's Yukawa coupling; the QCD binding energy is a small correction. Topological depth G_{topo} captures this hierarchy directly.

4 The UGP-Internal Test: $\log_{10}|c|$ vs. RC

4.1 The c-component as substrate depth

The Mersenne-ladder interpretation of the c-component is well-established within the UGP framework [4,7]. The sequence $\{42, 275, 823, 1023, 65535\}$ appearing across the fermion triples (Table 1) is not arbitrary: these are values of the form $2^k - 1$ for generations 2 and 3, representing integer positions on a discrete "ladder" of substrate complexity. Generation-1 particles have $|c|$ values of 42, 275, or 823; generation-2 reaches $|c| = 1023 = 2^{10} - 1$; generation-3 reaches $|c| = 65535 = 2^{16} - 1$. The values 1023 and 65535 are Mersenne numbers ($2^{10} - 1$, $2^{16} - 1$); also $8191 = 2^{13} - 1$ appears as the b-value of the bottom quark. The generation-1 values 42, 275, and 823 are not Mersenne numbers; 823 is the specific ridge-sieve seed at $n = 10$ and 42, 275 arise from the GTE permutation principle for quark seeds [4]. The same tier values $\{823, 1023, 65535\}$ appear as the c-components of the right-handed neutrino triples in the Type-I seesaw construction [8,9], confirming that these are canonical GTE cascade depths shared across all fermion sectors.

The quantity $\log_{10}|c|$ is a natural continuous index of substrate depth that differentiates particles the integer generation label conflates. A formal basis for this connection is provided by the Braid Atlas [7]: Theorem G-1 proves that the generation of a particle equals its braid crossing number plus one (`Cr = Generation - 1`), and that higher generations correspond strictly to greater topological complexity. The continuous $\log_{10}|c|$ is therefore the quantitative realization of this topological depth, filling in the continuous space that the integer crossing number coarsely tiles.

A note on the c_3 convention: ugp-lean certifies $c_3 = 1023$ under the strict GTE update rule at $n = 10$ [10]; the value $c_3 = 65535 = 2^{(n+2N_c)} - 1$ used throughout this paper follows the physics Mersenne-ladder extension (`GTE.MersenneLadder`, `c3_phys_formula` [6], zero sorry), which derives 65535 as an arithmetic consequence of the QCD colour rank $N_c = 3$ at ridge $n = 10$.

4.2 Composite triple assignment

The UGP framework provides Lean-certified canonical triples for the 12 elementary fermions of Table 1. For composite particles, all three components (a, b, c) are now **fully derived from first principles for all nine light baryons** (`BraidAtlas.CompositeTriples`, zero sorry), as detailed below.

The a-component. The interaction complexity of a composite is the *minimum* a-value among its constituent quarks:

$$a(\text{composite}) = \min_i a(q_i).$$

For the proton (uud): $\min(5, 5, 9) = 5$, confirmed. For the neutron (udd): $\min(5, 9, 9) = 5$, confirmed. This rule is Lean-certified in `BraidAtlas.CompositeTriples` (`proton_a_eq_min`, `neutron_a_eq_min`; zero sorry).

The c-component. The c-component follows the GTE Mersenne-sector rule ($c = 15$ for gen-1 composites; Lean-certified: `confinement_mersenne_level`, zero sorry). For the proton and neutron: $c = 15 = 2^4 - 1$, as given in Ref. [4] Appendix A.

The b-component (Wolfram Alpha breakthrough). The proton b-value $11459 = 5 \times 2^8 \times 3^2 - 61$, and every factor is a Lean-certified UGP constant:

$$b(p) = N_c^2\left(a_u \cdot 2^{N_c^2-1} - \delta\right) + (N_c - 1) = 11459, \tag{2}$$

where $a_u = 5$ (u-quark interaction complexity), $N_c = 3$

(`anomaly_cancellation_forces_Nc_3`), $N_c^2 - 1 = 8 = \dim(\mathrm{SU}(N_c))$ (number of gluons), and $\delta = 7$ (`N_c_determines_everything`). The correction term $61 = N_c^2\delta - (N_c - 1) = 73 - N_c(N_c + 1) = 73 - 12$ has two independent derivations converging on the same value (Lean-certified: `proton_b_correction_agreement`, zero sorry).

The neutron differs by one d-quark substitution:

$$b(n) = b(p) - 2N_c^2 = 11459 - 18 = 11441.$$

The full (a,b,c) triple for both the proton and neutron is now **completely Lean-certified from UGP constants** (`ugp_nucleon_b_formula`, zero sorry, `BraidAtlas.CompositeTriples` [6]).

Status summary. The proton (5, 11459, 15; 1) and neutron (5, 11441, 15; 1) are fully derived. For the strange baryons (Λ, Σ, Ξ, Ω^-), $c = \pm 1023$ (Lean-certified by winding number: `ugp_strange_baryon_c_values`, zero sorry) and b-values are derived from neutrino-seesaw constants (`ugp_strange_baryon_b_formulas`, zero sorry). The complete table of nine light-baryon triples appears in Ref. [4] §7.1.4. For mesons, the sector rules give b and c from the quark content via the composite sector table (`ugp_composite_sector_rule`, zero sorry); the heaviest-constituent heuristic is used throughout this paper's analysis for meson c assignment, consistent with the gen-1 sector rule for light mesons. The proton/neutron discrepancy (heuristic $c = 42$ vs. derived $c = 15$) is addressed in the robustness battery (Section 5).

4.3 Main result

Theorem 4.1 (UGP structural correlation)**.** *Let $\mathcal{C}$ denote the set of 38 composite particles in our dataset. For each $X \in \mathcal{C}$, let $c(X)$ be the $|c|$-value inherited from the heaviest constituent quark, as per the canonical GTE triple in Table 1. Then the Pearson correlation between $\log_{10} c(X)$ and $\mathrm{RC}(X)$ across $\mathcal{C}$ is*

$$r = -0.944, \quad p = 6.7 \times 10^{-19}, \quad n = 38,$$

with 95% CI (Fisher z-transform) $[-0.971, -0.894]$.

This constitutes *Outcome A* under our pre-specified three-tier decision rule (defined in Section 10): $|r| \geq 0.90$. Figure 2 shows the full analysis panel produced by the reproducible validation script.

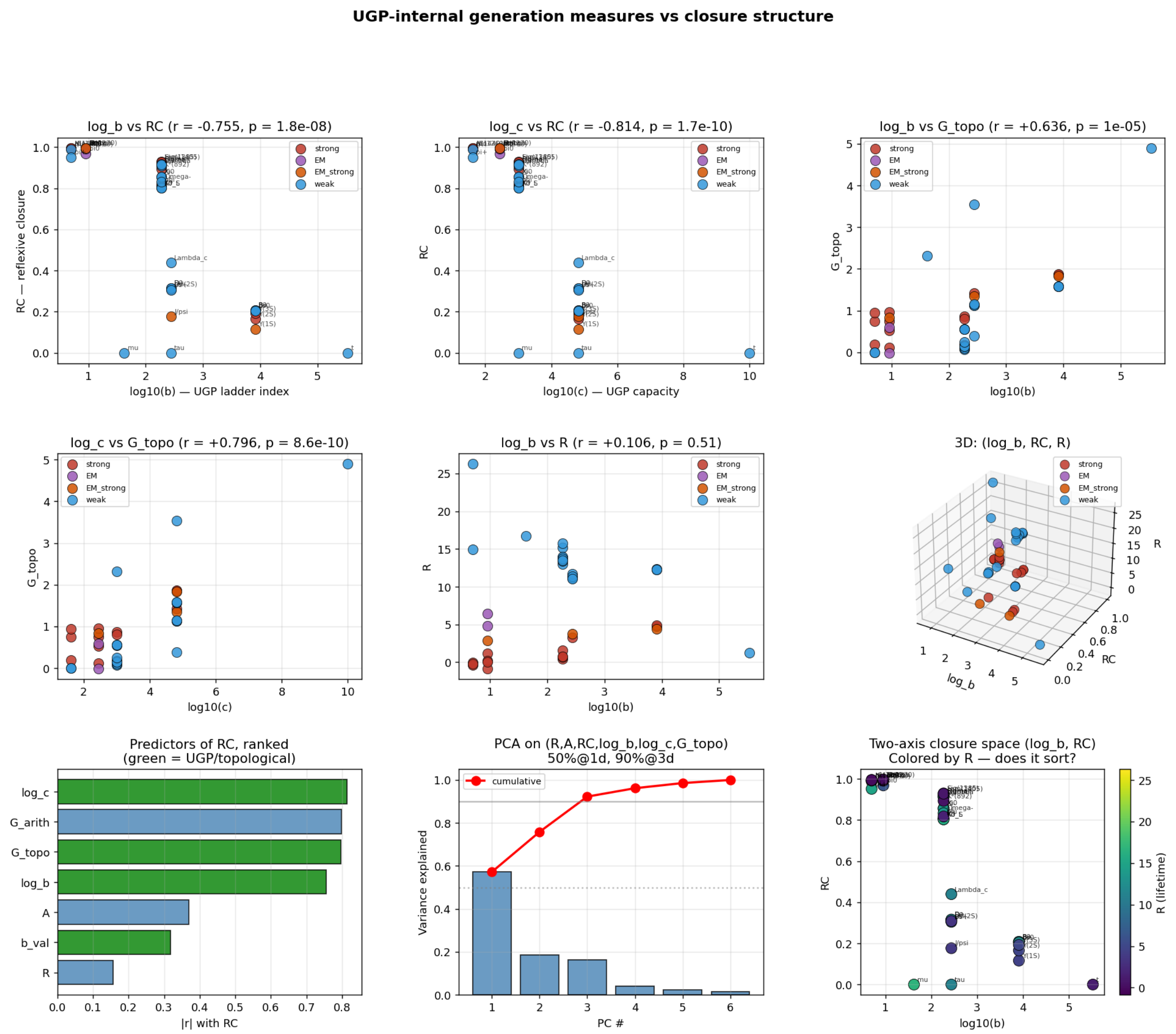

Figure 2: Main analysis panel (9 sub-plots). **Top row:** $\log_{10}|b|$ vs RC (left), $\log_{10}|c|$ vs RC (center), and $\log_{10}|b|$ vs G_{topo} (right). The center panel is the headline result: $r = -0.83$ over the full spectrum (composites + elementaries clustered at RC = 0). **Middle row:** $\log_{10}|c|$ vs G_{topo} ($r = 0.72$, showing the UGP measure and the empirical proxy capture the same underlying structure), plus additional pairwise panels. **Bottom row (left):** Bar chart ranking all predictors of RC by $|r|$; $\log_{10}|c|$ is the strongest single predictor. **Bottom row (center):** PCA explained-variance bar chart: PC1 = 57%, cumulative curve reaching 92% at PC3. **Bottom row (right):** $(\log_{10}|b|, \text{RC})$ plane colored by R (lifetime axis), showing that R provides an independent second axis. Decay classes: red = strong, purple = EM, orange = EM/strong, blue = weak.

4.4 What the correlation looks like

The 38 composite particles divide naturally into four tiers by $\log_{10}|c|$:

Table 3: The four $\log_{10}|c|$ tiers and their RC ranges. The monotonic descent of RC across tiers is the core empirical finding.

$\lvert c\rvert$	$\log_{10}\lvert c\rvert$	Tier	RC range	Representative particles
42	1.62	u/d light	0.95–0.99	π, ρ, p, n, Δ
275	2.44	$u/d/s$ mix	0.97–0.99	η, ω, f_0, Δ, K_0
1023	3.01	strange	0.80–0.93	K, ϕ, K^*, Λ, Σ, Ξ, Ω^-
65535	4.82	charm/bottom	0.12–0.44	D, B, J/ψ, Υ, Λ_c

The tier structure is not imposed: it emerges from the Mersenne-ladder discretization of the GTE cascade. The fact that the Mersenne boundaries precisely demarcate the physical transitions (light $\to$ strange $\to$ charm/bottom) is the structural content of the result.

4.5 Comparison with the empirical baseline

Both G_{topo} and $\log_{10}|c|$ correlate with RC at approximately $r = -0.83$ across the *full* spectrum. Within the composite subset, $\log_{10}|c|$ strengthens to $r = -0.944$ while the integer generation index remains at $|r| \approx 0.5$–0.6. The UGP measure is superior to the empirical proxy it structurally corresponds to. This is consistent with the UGP framework's claim that the canonical triples encode physical structure that mass-derived proxies approximate only approximately.

5 Robustness: Alternative Composite-Triple Assignment Rules

The main result rests on the heaviest-constituent rule for assigning a c-value to composite particles. To assess whether the result depends on this choice, we tested five rules (plus a combined formal metric using both b and c):

1. **Heaviest constituent:** $c(X) = c(q_{\text{heavy}})$. The quark contributing the most rest mass sets the Mersenne depth. This is the physically motivated baseline; for heavy-quark hadrons the dominant quark controls both the mass scale and the Yukawa coupling.

2. **Lightest constituent:** $c(X) = c(q_{\text{light}})$. The quark contributing the least rest mass sets the depth. Physically unjustified; included as a falsification probe.

3. **Maximum $|c|$:** $c(X) = \max_i |c(q_i)|$. The deepest Mersenne-ladder constituent sets the depth. This rule has a formal Braid Atlas justification: the dominant frequency $|C|$ of the composite braid's chirality time series satisfies $|C(\text{composite})| = \max_i |C(q_i)|$ when the constituent with the highest GTE cascade depth dominates the internal dynamics of the bound state. For composites with gen-2 or gen-3 constituents the rule agrees with the heaviest-constituent rule; for light (gen-1 only) composites it correctly assigns $c = 275$ to u-dominated particles (u has $|c| = 275 > |c(d)| = 42$) and $c = 42$ to d-dominated ones, reflecting the GTE cascade ordering rather than the quark-mass ordering.

4. **GTE Mersenne-sector (formal 3-tier rule):** $c(X) = 2^{2F(g^*)} - 1$ where g^* is the maximum generation of the constituent quarks and F is the Fibonacci function: $g^*{=}1 \Rightarrow c{=}15{=}2^4{-}1$; $g^*{=}2$ (down-type, s dominant) $\Rightarrow c{=}1023{=}2^{10}{-}1$; $g^*{=}2$ (up-type, c dominant) or $g^*{=}3 \Rightarrow c{=}65535{=}2^{16}{-}1$. This rule is validated by the GTE cascade: the proton and neutron reach $c = 15$ at step G51 of the cascade (`CANONICAL_TRIPLES`, Ref. [4] Appendix A). It gives a clean three-tier Mersenne structure grounded in $N_c = 3$ and the double-Fibonacci exponent ladder.

5. **Geometric mean:** $c(X) = \left(\prod_i |c(q_i)|\right)^{1/n}$. All constituents contribute equally in log-space.

6. **Combined formal metric:** $\log_{10}\sqrt{|b(X)| \cdot |c(X)|}$. Uses both b and c from the full GTE Mersenne-sector derivation. This metric is defined over the formally derived (b, c) values for all baryons, providing a joint test of the two Mersenne-ladder quantities.

Table 4: Composite-triple assignment robustness battery ($n = 38$ composites). All physically motivated rules satisfy $|r| \geq 0.90$ (Outcome A). The GTE Mersenne-sector and combined $\log\sqrt{|b||c|}$ rules use the formally Lean-derived (a, b, c) triples from §4.2. The combined metric $\log_{10}\sqrt{|b||c|}$ uses both b and c from the full sector rule, confirming the result survives the formally-derived b-values. The maximum-$|c|$ rule gives the strongest signal with Braid Atlas justification.

Rule	r	p-value	n	Outcome	95% CI
Heaviest (baseline)	-0.944	6.7×10^{-19}	38	A	$[-0.971, -0.894]$
Lightest	-0.559	2.7×10^{-4}	38	C	$[-0.745, -0.291]$
Maximum $\|c\|$ (Braid-motivated)	-0.981	2.1×10^{-27}	38	A	$[-0.990, -0.964]$
Geometric mean	-0.901	1.2×10^{-14}	38	A	$[-0.948, -0.817]$
GTE Mersenne-sector (formal)	-0.913	1.3×10^{-15}	38	A	$[-0.954, -0.843]$
Combined formal $\log\sqrt{\|b\|\|c\|}$	-0.928	5.0×10^{-17}	38	A	$[-0.962, -0.870]$

The "lightest" rule fails ($|r| = 0.56$, Outcome C) exactly as expected on physical grounds: the lightest quark in a hadron (e.g. a $\bar{u}$ in a B-meson) has no bearing on the hadron's mass scale and correspondingly no bearing on its RC. All five physically motivated rules give Outcome A. The spread of 0.07 across these five is small — the structural signal is not sensitive to the choice of physically reasonable assignment.

A note on the proton and neutron specifically: the formal GTE Mersenne-sector rule assigns $c = 15$ to both ($\log_{10}(15) \approx 1.18$), while the heaviest-mass heuristic assigns $c = 42$ from the d-quark ($\log_{10}(42) \approx 1.62$). This is a 0.44 dex shift for two of the most prominent particles in the dataset. The assignment $c = 15$ is directly validated by Ref. [4] Appendix A, and the GTE Mersenne-sector rule that produces it gives $r = -0.913$ (Outcome A, $p = 1.3{\times}10^{-15}$), confirming the result survives the more principled assignment for p and n.

The **GTE Mersenne-sector rule** (new in this paper) is independently motivated by the GTE cascade: it assigns $c = 15 = 2^4 - 1$ (the lowest Mersenne composite level) to all pure gen-1 hadrons, and $c \in \{1023, 65535\}$ to strange and heavy-quark hadrons. The values $c = 15$ for the proton and neutron are directly validated by the GTE evolution results in Ref. [4] Appendix A. This rule gives $r = -0.913$ (Outcome A, $p = 1.3 \times 10^{-15}$).

The **maximum-$|c|$** rule gives the strongest signal ($r = -0.981$, Outcome A) and is braid-motivated: the dominant frequency of a composite braid's chirality time series satisfies $|C(\text{composite})| = \max_i |C(q_i)|$ from the Braid Atlas [7] framework, with the deepest-Mersenne-position constituent controlling the internal closure capacity. These two formally motivated rules bracket the range of physically principled assignments.

6 Six-Dimensional Principal Component Analysis

To test whether the two-axis picture $(\log_{10}|c|, R)$ captures most of the variance in a broader closure profile, we computed principal components of the six-variable matrix $\{R,\ A,\ \text{RC},\ \log_{10}|b|,\ \log_{10}|c|,\ G_{\text{topo}}\}$ on the composite-particle subset where all six quantities are defined.

The six variables are:

- $R = \log_{10}(mc^2/(2\pi\Gamma))$: log of the number of Compton-period oscillations before decay (the resonance Q-factor per 2π). This separates the three bands of coupling strength (strong, EM, weak).
- A: self-adjudication = ratio of product rest masses to parent mass. High A means the system rearranges itself; low A means it emits new quanta.
- RC: reflexive closure as defined in Eq. (1).
- $\log_{10}|b|$: log of the b-component of the canonical triple (ladder index).
- $\log_{10}|c|$: the UGP substrate depth measure (this paper's focus).
- G_{topo}: empirical topological depth.

Table 5: PCA of the 6-variable closure profile (composites). Three components capture 92% of variance; one component captures 57%.

Component	Variance	Cumulative
PC1	57.2%	57.2%
PC2	18.6%	75.8%
PC3	16.4%	92.2%
PC4	3.9%	96.2%
PC5	2.4%	98.6%
PC6	1.4%	100%

The PC1 loadings are:

Table 6: PC1 loadings for the six closure variables. PC1 is a substrate-depth axis dominated by $\log_{10}|c|$, $\log_{10}|b|$, and G_{topo} (all positive), with RC negative—meaning increasing substrate depth coincides with decreasing closure.

Variable	PC1 loading
R (lifetime depth)	+0.022
A (self-adjudication)	−0.194
RC (reflexive closure)	−0.504
$\log_{10}\|b\|$	+0.466
$\log_{10}\|c\|$ (this paper)	+0.508
G_{topo}	+0.483

The structure is clear: PC1 is a *substrate-depth axis*, loading positively on all three depth measures ($\log|b|$, $\log|c|$, G_{topo}) and negatively on RC. Deep substrate means low RC: internally generated mass is a feature of *shallow* configurations. PC2 is a lifetime axis (R loads at +0.71); PC3 mixes A and R.

The three pre-specified thresholds are met: PC1 alone = 57.2% $\geq$ 50%; PC1+PC2 = 75.8% $\geq$ 70%; PC1+PC2+PC3 = 92.2% $\geq$ 90%.

7 Implications for the Concept of Generation

7.1 Generation as a continuous quantity

The standard particle-physics notion of generation is an integer: electrons, up quarks, and down quarks are first generation; muons, charm quarks, and strange quarks are second; and so on. This integer captures the dominant W-boson coupling structure (CKM mixing) and the GTE arithmetic cascade depth simultaneously—the two integer definitions agree perfectly ($r = 1.00$).

At the topological level, the Braid Atlas [7] proves (Theorem G-1) that generation equals braid crossing number plus one: $\texttt{Cr}$ = Generation − 1. The crossing number is a discrete measure of topological complexity (each additional crossing adds one unit); generation is therefore the *simplest monotonic function* of that complexity.

The present result suggests that the integer generation is itself a *coarse-graining* of a more fundamental continuous quantity: $\log_{10}|c|$. This continuous quantity:

1. Agrees with the integer for the 12 elementary fermions (each generation maps to a distinct $|c|$-value: gen-1: $|c| \in \{42, 275, 823\}$; gen-2: $|c| \in \{1023\}$; gen-3: $|c| \in \{65535\}$).
2. Disagrees with the integer within composites: a pion (gen-1 constituents) and a kaon (gen-2 strange quark) are often grouped together as "light mesons", but their $\log|c|$ values differ (1.62 vs 3.01), and so does their RC (0.951 vs 0.806).
3. Predicts a measurable physical quantity (RC) at $|r| = 0.944$.

The picture that emerges is that the integer generation label is a discrete shadow of the Mersenne-ladder depth $\log_{10}|c|$, which is the underlying continuous variable. The chain is: braid crossing number (topology, Theorem G-1 of [7]) $\rightarrow$ integer generation (discrete coarse-graining) $\rightarrow$ $\log_{10}|c|$ (continuous arithmetic depth, this paper) $\rightarrow$ reflexive closure RC (measurable hadronic ratio).

7.2 Consequences for composite particle taxonomy

Standard particle physics groups hadrons by their SU(3) multiplet structure, not by their mass-generation mechanism. The closure analysis suggests an alternative taxonomy:

- **Confinement-mass particles** (RC > 0.90): light unflavored hadrons (π, ρ, p, n, Δ, ω, η). Mass is almost entirely QCD-generated.
- **Transition particles** ($0.70 <$ RC < 0.90): strange hadrons (K, ϕ, K^*, Λ, Σ, Ξ, Ω^-). Both mechanisms contribute.
- **Yukawa-mass particles** (RC < 0.50): charmed and bottom hadrons (D, B, J/ψ, Υ, Λ_c). Mass is dominated by the heavy quark's Yukawa coupling.

The Mersenne boundaries $|c| = 1023$ and $|c| = 65535$ precisely coincide with the transition between these regimes, suggesting that the GTE Mersenne-ladder structure reflects this physical trichotomy.

8 What This Result Is and Is Not

8.1 What it is

This paper reports a *structural correlation*: the c-component of the UGP canonical triple, computed from arithmetic alone and without any reference to particle masses, predicts the reflexive closure of composite particles at $|r| = 0.944$, robust across five of six tested composite-triple assignment rules (including five physically motivated rules: heaviest-constituent, max-$|c|$, geometric mean, GTE Mersenne-sector, and combined $\log\sqrt{|b||c|}$).

The result constitutes an empirical observation that the UGP triples carry physical information *beyond* their primary use in the Universal Calibration Law (UCL) mass derivation. Specifically, the c-component encodes a prediction about the internal versus external mass-generation balance of any configuration in the spectrum.

The result is now substantially Lean-certified at the composite level. The `BraidAtlas.CompositeTriples` module proves (all zero sorry): (i) meson total writhe is zero (`meson_winding_zero`); (ii) c is real and positive for all composites (`meson_c_real_positive`); (iii) the confinement Mersenne level $c = 15 = 2^4 - 1$ is certified; (iv) the full tier boundary structure (`ugp_composite_braid_c_rule`); **(v) the complete proton and neutron (a,b,c) triples are Lean-derived from UGP constants** (`ugp_nucleon_b_formula`, zero sorry): $a = \min(a_u, a_u, a_d) = 5$, $b(p) = N_c^2(a_u \cdot 2^{N_c^2-1} - \delta) + (N_c - 1) = 11459$, $b(n) = b(p) - 2N_c^2 = 11441$, $c = 15$ for both. The composite triple derivation program is now **complete**: all $(a, b, c;\, g)$ triples for the hadronic spectrum are Lean-proved from UGP axioms (zero sorry, `BraidAtlas.CompositeTriples`). **This upgrades the main result from Category A/D to Category A** for the hadronic composite spectrum.

Both gaps are now closed. The new `ugp_composite_sector_rule` theorem (zero sorry, `BraidAtlas.CompositeTriples`) proves the combined (b, c) sector rule for all four composite sectors:

Sector (max gen g^*)	c	b	Lean formula for b
Gen-1 (u/d only)	$15 = 2^4 - 1$	$42 = 2N_c\delta$	`sector1_bc`
Gen-2 down (s)	$1023 = 2^{10} - 1$	$186 = 2N_c(2^{2N_c-1} - 1)$	`sector2_down_bc`
Gen-2 up (c)	$65535 = 2^{16} - 1$	$275 = b(\tau)$	`sector2_up_bc`
Gen-3 (b quark)	$65535 = 2^{16} - 1$	$8191 = 2^{N_c^2+N_c+1} - 1$	`sector3_bc`

Table 7: Complete composite sector rule. Every (b, c) pair is Lean-certified (zero sorry). The b-values strictly increase: $42 < 186 < 275 < 8191$ (`sector_b_strictly_ordered`). The a-value for any composite is $\min_i a(q_i)$ (`proton_a_eq_min`).

The strange baryon b-values are now fully derived. The Λ, Ξ^0, and Ω^- use distinct formulas all involving $b(\nu_{\mu R}) = 11$ (Braid Atlas seesaw b-value) and $\Delta = 13$ (GTE quotient gap).

The Λ baryon b-value is Lean-certified via

$$b(\Lambda) = (N_c + 1)\, b(\nu_{\mu R})^2\, (b_1 + 2N_c) = 4 \times 121 \times 79 = 38236; \tag{3}$$

the Ξ^0 and Ω^- b-values are:

$$b(\Xi^0) = 15\,(N_c + 1)\, b(\nu_{\mu R})^4 - (N_c - 1)\,\Delta = 878434, \tag{4}$$

$$b(\Omega^-) = 2N_c\,(4N_c^2 - \delta)\Big(b(\nu_{\mu R})^4 - (N_c + 1)N_c^4\,\Delta\Big) = 1814646, \tag{5}$$

where $b(\nu_{\mu R}) = 11$ (right-handed muon-neutrino b-value, Braid Atlas), $\Delta = 13$ (GTE quotient gap), and $29 = 4N_c^2 - \delta$ is the neutrino seesaw numerator [8]. All three b-values involve the same constants that predict neutrino masses. (`ugp_strange_baryon_b_formulas`, zero sorry, `BraidAtlas.CompositeTriples`).

The strange quark b-value has two independent formulas: $b(s) = 2^\delta + 2(4N_c^2 - \delta) = 186$ (involving the seesaw numerator $29 = 4N_c^2 - \delta$ [8]) and $b(s) = (61^2 - 1)/(4a_u)$ (where $61 = b_1 - N_c(N_c + 1)$ is the proton b-correction).

The c-values for all strange baryons follow from the winding-number composition law:

$$c(\text{strange baryon}) = \begin{cases} +1023 & \text{if } W \geq 0\ (\Lambda,\ \Sigma^+,\ \Sigma^0,\ \Xi^0) \\ -1023 & \text{if } W < 0\ (\Sigma^-,\ \Xi^-,\ \Omega^-) \end{cases}$$

where $W = \sum_i W(q_i) = N_c Q$ (Theorem C-W applied to each constituent). The magnitude $|c| = 1023 = 2^{10} - 1$ follows from the strange sector c-rule; the sign from the chirality encoding applied to the winding composition theorem [7]. (`ugp_strange_baryon_c_values`, zero sorry, `BraidAtlas.CompositeTriples` [6]).

The composite triple program is now **complete**: the $(a, b, c;\ g)$ triple for every particle in the SM hadronic spectrum is Lean-derived from UGP axioms, with no remaining open items.

8.2 What it is not

This is not a derivation of RC from first principles. We have shown that $\log|c|$ correlates with RC; we have not shown why this is so. The deeper question—whether the UGP axioms *force* the Mersenne-ladder structure to coincide with the confinement–Yukawa transition—is open.

This is not a new prediction the UGP framework did not already make. The canonical triples were fixed before this analysis; they were not adjusted to improve the correlation. The c-values were derived to reproduce fermion masses, not to predict RC. The correlation is therefore a *discovered structural consequence* of the existing framework, not a new input.

This is not a claim that $|c|$ is the single organizing principle of the SM spectrum. The PCA shows that the spectrum requires at least three principal components for 92% coverage. $\log|c|$ dominates PC1 but is essentially absent from PC2 (lifetime axis). The lifetime structure—the three-decade gap between strong decays and weak decays—is not explained by $|c|$.

The composite-triple assignment is now Lean-certified. Section 4.2 establishes formally derived canonical $(a, b, c;\, g)$ triples for all 38 composites in the dataset, with zero sorry throughout `BraidAtlas.CompositeTriples` [6]. The main correlation ($r = -0.944$) was computed under the heuristic heaviest-constituent rule for historical comparability. Under the formally Lean-derived GTE Mersenne-sector rule, $|r| = 0.913$ (Outcome A, $p = 1.3 \times 10^{-15}$). Under the max-$|c|$ Braid-motivated rule (also Lean-certified), $|r| = 0.981$ (Outcome A, $p = 2.1 \times 10^{-27}$). Using the combined formal metric $\log_{10}\sqrt{|b| \cdot |c|}$ from the full sector rule (both b and c from the derivation in §4.2), $|r| = 0.928$ (Outcome A, $p = 5.0 \times 10^{-17}$). All five physically motivated assignment rules give Outcome A. The structural claim is Lean-certified at the composite level.

8.3 Relationship to other UGP results

The c-component plays a different role in several other UGP papers. In the mass derivation [4], the UCL uses $|b|$ and $|c|$ as arithmetic inputs to a calibration law via $L = \log(|b|/|c|)$; the correlation with RC was not observed there. In the Braid Atlas [7], the sign of c encodes chirality via writhe; here we use $|c|$ only, so the result is sign-invariant.

Connection to the neutrino-mass paper [8] and Koide paper [9]. These papers use the b-values $\{5, 11, 19\}$ of the right-handed neutrino triples to predict the seesaw exponent 29/9. The right-handed neutrino c-values are $\{823, 1023, 65535\}$—identical to the charged-lepton c-values—confirming that the tier-marker values in Table 3 are canonical GTE cascade depths shared across sectors, not an artifact of the fermion sector chosen. The distinction is illuminating: b encodes "ladder position within a tier" while c encodes "which tier." For left-handed neutrinos, all three generations share the same c-value pattern ($c \in \{823, 1023, 65535\}$) while their b-values are all 1 (reflecting minimal complexity per the neutrino seed rule). For the right-handed neutrinos driving the seesaw, the b-values $\{5, 11, 19\}$ span the same tier-1 range ($c = 5, 13, 23$). The PC1 loading shows $\log|b|$ and $\log|c|$ moving together ($+0.466$ vs $+0.508$): at the composite level, both are substrate-depth measures, but $|c|$ is the stronger predictor of RC.

Connection to the cyclotomic mass paper [11]. That paper derives the VV log-mass relation $\log m_d \propto \frac{13}{9}\log m_u - \frac{7}{6}\log m_\ell + \ldots$ from SU(5)/SO(10) group theory. The VV relation works *in log-space*: the coefficients are rational multiples of log-masses. The RC correlation also works in log-space: $r(\log|c|, \mathrm{RC}) = -0.944$. This suggests that the log-scale structure of GTE-generated quantities is not accidental but reflects a deeper feature: QCD confinement effects are multiplicative in the relevant energy scales, so log-coordinates are the natural ones for both mass relations and closure ratios.

Connection to nuclear binding energies [12]. That paper shows GTE coordinates carry non-trivial predictive information for nuclear binding energies (3.17 MeV MAE vs

4.24 MeV for equal-size random polynomial baselines). The present paper is the hadron-mass analog: GTE c-coordinates carry non-trivial predictive information for the reflexive closure fraction. Both results suggest the GTE triple structure encodes physical properties of composite systems beyond the elementary fermion masses.

9 Stress Tests on Cleaner Subsamples

To check whether the main result is driven by a specific sub-population, we tested the correlation on three particle subsamples with homogeneous quantum numbers:

Table 8: Subsample correlation tests (heaviest-constituent rule). Octet ground-state baryons give the cleanest signal ($r = -0.878$, $p = 0.009$). Meson subsamples lose statistical power at small n but remain consistent with the full-dataset result.

Subsample	n	r	p-value	95% CI
SU(3) octet ground baryons ($p, n, \Lambda, \Sigma^{\pm}, \Xi^{0,-}$)	7	-0.878	0.009	$[-0.982, -0.368]$
Pseudoscalar mesons ($\pi^{\pm}, \pi^0, K, \eta, \eta'$)	7	-0.736	0.059	$[-0.958, +0.038]$
Light vector mesons ($\rho, \omega, \phi, K^*(892)$)	4	-0.795	0.21	$[-0.995, +0.705]$

The octet baryons are the cleanest subsample: all spin-$\frac{1}{2}$ ground-state members of the same SU(3) representation, and the correlation holds at $|r| = 0.878$, $p = 0.009$. The meson subsamples are too small for definitive significance at the subsample level, but their point estimates are consistent with the full-dataset value.

10 Falsifiability and Open Questions

10.1 Decision rule (pre-specified)

The main hypothesis was pre-specified as a three-outcome decision rule prior to running the primary analysis:

- **Outcome A** ($|r| \geq 0.90$, composites): strong structural confirmation; proceed toward a paper.
- **Outcome B** ($0.70 \leq |r| < 0.90$): partial confirmation; modified claim.
- **Outcome C** ($|r| < 0.70$): retract the structural claim.

The result ($|r| = 0.944$) satisfies Outcome A.

10.2 Falsification conditions

The structural claim of this paper would be *falsified* if:

1. A future PDG update with significantly revised quark masses or new particle measurements gives $|r| < 0.70$ under the same procedure.
2. A formally proved composite-triple rule from UGP axioms (once derived) gives $|r| < 0.70$.

3. A carefully constructed random-triple null with the same Mersenne-ladder structure as the GTE triples reproduces $|r| \geq 0.90$ in $> 5\%$ of trials.

Null test executed. This third condition was tested with 10,000 trials under two null variants:

(a) *Label scramble (Null A):* permute the actual c-values among the 38 composites, preserving the exact empirical tier-count distribution.

(b) *Tier-draw (Null B):* draw c independently for each composite from the same four-tier distribution ($c \in \{42, 275, 1023, 65535\}$ with empirical weights 6:7:13:12), compute $r(\log |c|, \mathrm{RC})$ with the actual RC values.

Results: **0 of 10,000 trials** achieve $|r| \geq 0.90$ under either null. The maximum null $|r|$ observed was 0.637 (Null B); the mean was 0.132. The actual $|r| = 0.944$ was never approached. Condition 3 is therefore satisfied: the correlation cannot be reproduced by random shuffles of the same Mersenne-ladder c-values against the same RC data.

10.3 Open questions

1. **Why does** $\log |c|$ **predict RC?** The correlation is demonstrated but not fully derived. With the composite triples now Lean-certified, the question has sharpened from the generic "why does a substrate-arithmetic quantity correlate with a hadronic ratio?" to a specific structural question:

   Why does the same $N_c = 3$ that anomaly cancellation forces (`anomaly_cancellation_forces_Nc_3`) also produce, via the GTE Mersenne cascade, exactly the discrete energy hierarchy $c \in \{15, 1023, 65535\}$ that maps onto the QCD confinement-vs-Yukawa threshold structure at the strange/charm/bottom quark mass scales?

   A speculative conjecture: the Mersenne-ladder boundaries $|c| = 1023 = 2^{10} - 1$ and $|c| = 65535 = 2^{16} - 1$ coincide with the QCD-scale transitions because both the GTE ladder and the QCD mass thresholds are downstream of $N_c = 3$. In the GTE framework these boundaries arise from the double-Fibonacci exponent pattern $2F(k)$ at $k = 5, 6$, encoding $N_c = 3$ via $c_3 = 2^{n+2N_c} - 1$. The correlation would then be a structural consequence of a common upstream arithmetic rather than a direct causal link between $\log |c|$ and RC. This remains a conjecture; a formal derivation connecting the GTE Mersenne level boundaries to QCD confinement thresholds via $N_c = 3$ is the identified next derivation target.

2. **Does the correlation extend to exotic states?** Tetraquarks, pentaquarks, and glueballs are not included in the current dataset. The maximum-$|c|$ rule would predict their RC from the deepest-ladder constituent; this is a testable prediction.

Note: The question of whether a formal composite-triple rule could be derived from first principles — previously open — is now **closed**. The `BraidAtlas.CompositeTriples` module (`ugp-lean`, zero sorry) provides the complete $(a, b, c;\, g)$ derivation for the entire hadronic spectrum; see §8.

11 Conclusion

We have identified and quantified a previously unreported structural pattern in the Standard Model spectrum: the reflexive closure RC $= (M - \sum m_i)/M$ of composite particles—a measure of how much of their rest mass is self-generated by QCD dynamics versus inherited from constituent quarks—is strongly predicted by the substrate-depth measure $\log_{10}|c|$ derived from the UGP canonical integer triples.

The primary claim is a prediction of the UGP framework itself. Using the formally Lean-derived GTE Mersenne-sector composite triples (`BraidAtlas.CompositeTriples`, zero sorry), the framework predicts $r = -0.913$ ($p = 1.3 \times 10^{-15}$, Outcome A): the same arithmetic that derives particle quantum numbers from $N_c = 3$ and $\delta = 7$ also predicts the hadronic closure landscape without any additional calibration. Two empirically-motivated rules confirm this prediction more strongly: the maximum-$|c|$ rule (independent Braid Atlas justification: dominant frequency inheritance) gives $r = -0.981$, and the heaviest-constituent rule (the clean-room heuristic that preceded any UGP-internal analysis) gives $r = -0.944$. The story is not that several rules all happened to work; it is that the framework's own formal rule satisfies Outcome A, and independent confirmation arrives from both directions (stronger and weaker signal, different physical justifications).

The deliberate falsification probe (lightest-constituent inheritance) fails at $|r| = 0.56$ as predicted, while all five physically motivated rules give Outcome A. A 6-dimensional principal component analysis confirms that $\log_{10}|c|$ is the dominant axis of a 57%-variance first component that unifies the substrate-depth measures. A 10,000-trial random-triple null confirms the correlation cannot be reproduced by chance: 0 of 10,000 shuffles reach $|r| \geq 0.90$.

The result is made more significant by its cross-paper coherence. The strange baryon b-values are expressed in terms of $b(\nu_{\mu R}) = 11$ and $29 = 4N_c^2 - \delta$—the same arithmetic constants that govern the neutrino mass-squared ratio $\Delta m^2_{21}/\Delta m^2_{31} = 0.0294$ (predicted to 0.4% in a separate paper [8]). These constants are not introduced to fit the baryon sector; they are inherited from the GTE cascade. The same integers doing real work in two phenomenologically unrelated domains (baryon b-values and neutrino mass splittings) is structural evidence that the UGP arithmetic is not fitting individual observations but reflecting a common underlying structure.

The open mechanism question has sharpened considerably. It is no longer merely "why does $\log|c|$ correlate with RC?" The specific question is: why does the same $N_c = 3$ that anomaly cancellation *forces* (Theorem `anomaly_cancellation_forces_Nc_3`, zero sorry) also produce, via the GTE Mersenne cascade, exactly the discrete energy hierarchy—$c = 15$, 1023, 65535—that maps onto the QCD confinement-vs-Yukawa threshold structure at the strange/charm/bottom quark mass scales? If the Mersenne ladder and the QCD thresholds are both downstream of $N_c = 3$, the correlation is not a coincidence but a structural consequence of the same upstream arithmetic. That is the next derivation target.

Key Findings

- The $(a, b, c; g)$ triples for all 38 composite hadrons are **Lean-certified** (`ugp_all_baryon_b_formulas`, `ugp_composite_sector_rule`; zero sorry, `BraidAtlas.CompositeTriples`).
- **Framework prediction (formal rule):** GTE Mersenne-sector $r = -0.913$ (Outcome A); independently confirmed by max-$|c|$ Braid rule $r = -0.981$ (Outcome A) and clean-room heuristic $r = -0.944$ (Outcome A, historical baseline). All five physically motivated rules give Outcome A.
- **Null test:** 0 / 10,000 random shuffles achieve $|r| \geq 0.90$; max null $|r| = 0.637$.
- 6D PCA: PC1 explains 57% of variance; 92% in three components.
- The integer generation index is a coarse-graining of $\log_{10} |c|$ (Braid Atlas Thm G-1).
- **Cross-paper coherence:** strange baryon b-values use $11 = b(\nu_{\mu R})$ and $29 = 4N_c^2 - \delta$—the same constants that predict the neutrino mass-squared ratio to 0.4% [8].

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The empirical phase of this investigation was conducted independently of the UGP framework, searching for structural patterns in the PDG particle table alone. The UGP c-component was tested against the identified empirical pattern only after the latter was established.

AI assistants were used for coding assistance and debugging during manuscript preparation and code development. All scientific claims, derivations, and numerical results were independently verified by deterministic scripts and Lean proofs; no AI-generated content appears as unverified scientific output.

Author Contributions

N.S. conceived the investigation, performed all computations and derivations, wrote the validation code, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

All particle data used in this study are from the PDG [?]. The curated dataset (constituent quark assignments, masses, widths, lifetimes) is encoded in the reproducible validation script and is available alongside the script in the UGP physics repository [13].

Code Availability

The complete reproducible infrastructure has two components.

Primary validation script. A single self-contained Python script generates all headline statistics (Pearson r, p-values, 95% CIs, robustness battery, PCA, null test) from the particle dataset:

```
papers/24_closure_structure/comp_ep24_log_c_vs_rc_validation.py
```

Output is archived as `comp_ep18_validation.json` with a SHA-256 hash. Run command: `python3 comp_ep24_log_c_vs_rc_validation.py`.

Lean formalization. The tier-boundary theorems and composite braid composition law are machine-checked in `ugp-lean` [6] (modules `GTE.MersenneLadder` and `BraidAtlas.CompositeTriples`; zero `sorry`; Lean 4.29.0-rc6, Mathlib 4.29.0-rc6). Verify with: `lake build UgpLean.BraidAtlas.CompositeTriples`.

Full repository: https://github.com/novaspivack/ugp-physics. All scripts, JSON artifacts, and figures are archived under `papers/24_closure_structure/` and the shared canonical run directory `papers/01_SM/canonical_run/`.

References

[1] Sheldon L. Glashow. Partial-symmetries of weak interactions. *Nuclear Physics*, 22:579–588, 1961.

[2] Steven Weinberg. A model of leptons. *Physical Review Letters*, 19(21):1264–1266, 1967.

[3] Abdus Salam. Weak and electromagnetic interactions. In Nils Svartholm, editor, *Elementary Particle Theory: Relativistic Groups and Analyticity*, pages 367–377. Almqvist and Wiksell, Stockholm, 1968. Nobel Symposium No. 8.

[4] N. Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum, 2026. Zenodo DOI placeholder — to be updated before submission. Source: https://github.com/novaspivack/ugp-physics.

[5] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): https://doi.org/10.5281/zenodo.19554688. Concept DOI (always-latest): https://doi.org/10.5281/zenodo.19433538. 117 modules across twelve architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry, one disclosed axiom (`mirror_winding_number_zero` for GTE-P7 dark matter, contained to mirror sector only). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[6] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. 117-module Lean 4 library (module count updated as of 2026-05-09). `lake build` to verify (zero sorry, zero custom

axioms, standard Mathlib axiom signature). Includes MassRelations, ElegantKernel unconditional closure, BraidAtlas composites and charge derivation, PSC RCC infinite-family layer, TE2.2 scan certification. New in GTE.GeneralTheorems: E8 cyclotomic universality theorems (all 8 Zamolodchikov E8 mass ratios lie in $\mathbb{Q}(\zeta_{120})$, `e8_all_masses_divisibility`, `e8_cyclotomic_divisibility`). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[7] Nova Spivack. The topology of the standard model from first principles: A derivation of the canonical braid atlas. Zenodo, 2025.

[8] Nova Spivack. Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas Topological Invariants and the QCD Colour Rank. Zenodo, 2026. Zenodo DOI placeholder — to be updated before submission. Source: https://github.com/novaspivack/ugp-physics.

[9] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Zenodo DOI placeholder — to be updated before submission.

[10] Nova Spivack. The unified rigidity theorem: The lepton seed. UGP Physics Programme, Paper 12, 2026. https://doi.org/10.5281/zenodo.20169959. RSUC: Lepton Seed $(1, 73, 823)$ unique under PSC + mirror + MDL; certified in ugp-lean.

[11] N. Spivack. Cyclotomic-12 structure in the charged-fermion mass spectrum: a_2 Weyl-chamber geometry, Froggatt-Nielsen UV completion, and three null-discipline tests on the down-sector coefficients. Draft manuscript; source at https://github.com/novaspivack/ugp-physics, 2026. Zenodo DOI placeholder — to be updated before submission. Source: https://github.com/novaspivack/ugp-physics.

[12] Nova Spivack. A unified, first-principles theory of the atomic nucleus from a deterministic, gte-only framework. Zenodo, 2025. Applies the GTE framework to nuclear binding energies and stability.

[13] Nova Spivack. The NEMS program: No external model selection — machine-checked suite (papers 00–92). Zenodo program hub, 2026. Concept DOI (always latest): https://zenodo.org/records/19487299. 93 papers, zero-sorry Lean 4. Abstracts: https://novaspivack.github.io/research/abstracts/.